

DSA STUDY BLUEPRINT SERIES

67. Largest Rectangle in Histogram Best Solution

Calculating Largest Rectangular Area in Histograms

Section 08 • C++ Core Data Structures & Algorithms

Core Concepts & Visual Blueprint

Theoretical models and memory architectures

Key Principles

- **Area Bounds:** Area is determined by height and distance between first smaller boundaries on left & right.
- Compute previous smaller and next smaller indices using monotonic stacks.

Memory & Execution Blueprint

Histogram Bars: [2, 1, 5, 6, 2, 3] $\Rightarrow$ Largest Area = **10**
(using 5 and 6, width = 2)

C++ Implementation Reference

Standard code layout and syntax

```
// Find prev smaller and next smaller indices for all elements.  
// Max Area = max(heights[i] * (nextSmaller[i] - prevSmaller[i] - 1)).
```

Algorithmic Complexity & Best Practices

Performance evaluations and critical guidelines

Performance Table

Aspect / Case	Complexity
Time Complexity	$O(N)$
Space Complexity	$O(N)$

Key Practices & Gotchas

- **Important:** Can also be implemented in a single pass using a stack that tracks boundary indices.
- Verify boundary conditions (e.g. empty inputs, single elements).
- Optimize dynamic memory allocations to prevent leaks.

Dry Run & Step-by-Step Execution

Trace analysis on a sample input case

Execution Walkthrough

Let's trace monotonic stack modifications for next greater element on `[4, 5, 2]` :

Active Element	Stack State (Indices)	Pop comparison	Result output
4	[0] (value 4)	N/A	Waiting for next greater
5	[1] (value 5)	5 > 4 (pop index 0)	Next Greater of 4 is 5

Interview Variations & Optimization Pathways

Common follow-up problems and optimization strategies

Key Variations & Follow-up Questions

- **Monotonic properties:** Stacks process elements strictly increasing or decreasing, optimizing nested lookups.
- **Min Stack details:** Track current minimum bounds by storing elements as encoded difference values.
- **Related LeetCode Problems:** #155 Min Stack, #84 Largest Rectangle in Histogram, #42 Trapping Rain Water.